Smart sensors now have wireless capabilities, allowing for their optimal placement around the home. Many smart sensors have the requisite intelligence to ascertain what a true positive-alarm trigger condition is, thus avoiding false positives. Programming options through improved command and control products, like microprocessors, allow for custom tailoring of sensors, especially for such specifications as sensitivity and multiple confirmation triggers that assist with improving accuracy in identifying a legitimate alarm trigger.

Occupant health monitoring is an expanding area of development with the creation of powerful accelerometer monitors and motion/temperature sensors. Sensors have expanded to include monitoring water leaks, carbon-dioxide and carbon-monoxide buildup, and other volatile organic compounds (VOCs). Likewise, other important items that concentrate on the health and efficiency of systems such as HVAC performance relative to specifications undergo monitoring as well.

Fig. 2: Automated security entry systems on the gatepost of a house offer an additional set of eyes for a homeowner's protection

Power

Turning our attention to the four key technology-related design considerations of power, sensing, command and control, and communications, which enable home automation security and monitoring, we will start by examining how power management technology advancements are assisting in this effort.

Power management is a key design challenge in every Internet of Things (IoT) and home automation security product. The ability to answer high peak current demands and very lengthy, low-power system standby times requires solutions that extend battery life. Electronic components manufacturers focus on offering a wide variety of products in which electrical energy gets efficient utilisation and distribution where it is essential.

Non-isolated direct current to direct current (DC/DC) and point-of-load (POL) power supplies function in security and monitoring applications. On input side, power may come in via wired-power options, often through 12V DC or using a battery-powered option, with the possible inclusion of supercapacitors—a type of high-capacity capacitor—or often in the form of electric double-layer capacitors (EDLCs).

DC/DC power converters that adjust input voltage to what the system needs to output are often in operation. Designers working on such power converters may consider high-quality, synchronous buck converters, as these are compact and easy to use. These parts may provide ample efficiency advantages over low-dropout (LDO) regulators.

When home automation systems utilise battery-powered devices, often designers employ step-down converters. These parts assist with battery-based applications, as these are capable of drawing very low quiescent currents (IQs), guaranteeing that the parts will draw low currents from the battery when in a wait state.

Capacitors are key components for efficient power conversion. One common thread among high-efficiency switch-mode power supply, microprocessor and digital circuit applications is the need to reduce noise while operating at higher frequencies. Specifically valuable for filtering, tantalum capacitor technology has many of the ideal characteristics that DC/DC converters, power supplies and other applications require. In home automation applications, power supply capacitors should exhibit many of the following positive characteristics:

- High capacitance retention at high frequencies
- Low failure rate
- Wide voltage range
- Surge robustness
- Environment (moisture/temperature) resistance
- Low cost

Power designs require protection against transient voltages. Using a capacitor is a common technique to control transients and protect the circuit in voltage control designs. A capacitor may operate across the line/pin to integrate the voltage and assist in the prevention of electrostatic discharge (ESD) damage. Engineers often over-estimate the performance of a standard multilayer ceramic capacitor (MLCC) because of the capacitor's significant value drop upon an applied ESD event. Introduction of supercapacitors, in the place of MLCCs, will actually make a design more robust.